

PhytoGraph Track 2 / M1.4 — Paleobotany & Megafauna Substrate Ingestion (clone 2, cycles 1-3)

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[OUTPUT: report]

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Abstract

Clone 2 of fanout e34b5b2c1c6c was scoped to milestone **M1.4** of the PhytoGraph campaign: stage paleobotany and megafauna specialty sources for Track 2 (Ghost Hyperedges). The required artifact — `substrate/staging/paleobotany_sources/INGEST_AUDIT.md` — was produced in cycle 1, alongside per-source staging tables for PBDB, the Late Quaternary Extinctions database (LQE), Faurby & Svenning megafauna ranges, and IUCN extant-disperser scaffolding. Targets were met: 237 extinct-fauna nodes (target ≥ 200), 31 literature-cited anachronism candidate edges (zero inferred), 52 distribution edges with verbatim source confidence codes, schema-conformance tests 10/10 passing. Cycles 2 and 3 were structural null cycles: the brief, worker, and auditor all converged on a PIVOT-to-exit posture, holding the boundary against scope creep into adjacent clones’ namespaces (Barrier 1, sibling staging, root deliverables) and awaiting harness-level termination. The merge report at the conductor handoff path is intact and unchanged since cycle 1.

1. Introduction

PhytoGraph is a typed hypergraph of plant biology used as a discovery instrument across six tracks. Track 2 (Ghost Hyperedges) tests the Janzen–Martin “neotropical anachronism” hypothesis at scale by looking for fruit syndromes whose modern disperser is missing. That test requires a substrate layer of extinct megafauna with date and range, modern dispersers with overlapping range, and paleoclimate context. Wave 1 of the campaign assigns this ingestion to milestone **M1.4** and fans it out to a dedicated clone — this clone.

The clone’s directive was narrow and disciplined:

- Stage from four specialty sources (PBDB, LQE, Faurby & Svenning, IUCN).
- Produce `extinct_fauna` nodes, `paleoclimate_paleoecology_context` nodes, megafauna range polygons, and extant-disperser nodes.
- ≥ 200 extinct-fauna nodes with date + range.
- Carry **source-stated confidence** on every record; do not inflate interpretive paleobotany claims.
- Stage `anachronism_candidate_edges` **only** where the source explicitly asserts one; **do not infer** them this cycle.
- Write to per-source staging tables; do not modify the shared substrate (that is Barrier 1's job).

This report covers cycles 1 through 3 of the clone's life: one substantive cycle followed by two null cycles whose audits independently confirmed scope exhaustion.

2. Approach

2.1 Per-source staging layout

Each source got its own directory under `substrate/staging/paleobotany_sources/`, isolated to prevent cross-source contamination before Barrier 1:

```
paleobotany_sources/
pbdb/                PBDB extinct fauna + paleo context
lqe/                 Late Quaternary Extinctions + source citations
faurby_svenning/     Megafauna range nodes + distribution edges + range polygons
iucn/                Extant-disperser scaffold
anachronism_canon/   31 literature-cited Janzen-Martin canon edges
_lib/                shared schema helpers
tests/               schema-conformance test suite
QUARANTINE/          reserved for non-conformant records (0 used)
INGEST_AUDIT.md       required output artifact (321 lines)
coverage_row.json    machine-readable coverage summary
```

Each per-source directory carries the build script that produced its `.jsonl` tables, so the ingest is reproducible row-for-row when live network access is restored.

2.2 Schema discipline

All staged records conform to PhytoGraph schema v1.0 (frozen at Barrier 0). Confidence is recorded verbatim from each source rather than re-coded into a unified scale, preserving interpretive provenance:

- **PBDB:** record-level taxonomic confidence + stratigraphic interval.
- **LQE:** extinction-date interval and certainty flag as published.
- **Faurby & Svenning:** `range_type_code` preserved verbatim (e.g. `present-natural` vs. `current`), which is the field downstream consumers need to distinguish reconstructed Late Pleistocene range from anthropogenically-shrunk modern range.
- **Anachronism canon:** every edge cites a published source asserting the anachronism claim (Janzen & Martin 1982 and successor literature); zero inferred edges.

2.3 Hold-out reservation

The 31 literature-cited anachronism edges in `anachronism_canon/` are explicitly reserved as the Track 2 held-out validation set for milestones M2.T2 and M3.T2. The merge report flags this

so the downstream ghost-partner ranker declares no-leakage upfront — the canon must not be fed into the model and then “recovered.”

3. Results

3.1 Cycle 1 — substantive ingestion

Cycle 1 produced the full M1.4 deliverable. The audit verified the following against the directive:

Target	Required	Delivered
extinct_fauna nodes with date + range	≥200	237
Literature-cited anachronism edges	≥0 (no inference)	31 (zero inferred)
Megafauna distribution edges	—	52 (verbatim range_type_code)
Schema-conformance tests	passing	10/10
Quarantined records	—	0
Raw IUCN polygons committed	—	0 (scaffold only; live download deferred)
INGEST_AUDIT.md	required	present, 321 lines
merge_report.md at conductor path	required	present

Live network access for the IUCN polygon download and a live PBDB re-query was unavailable during the cycle; the build scripts capture the canonical query keys so a row-for-row re-ingest is possible at Barrier 1. This limitation is documented in INGEST_AUDIT.md and surfaced in the merge report.

The cycle-1 audit issued a VALIDATED verdict on M1.4. No CRITICAL or MODERATE findings were filed against the staging output.

3.2 Cycle 2 — first null cycle

The cycle-2 brief, recognizing that M1.4 was already validated and the clone’s scope was exhausted, instructed zero further work. The worker complied: zero filesystem mutations. The auditor invoked the <no-null-cycle-validation> framework rule — a cycle producing only restatement-of-prior-state cannot itself be VALIDATED — and issued a **PIVOT-to-exit** verdict. One MINOR finding was logged: the harness’s low-output detector had not yet terminated the clone loop. This finding is harness-level, not agent-level; no agent action resolves it.

3.3 Cycle 3 — second null cycle

Cycle 3 was identical in posture to cycle 2: zero mutations, the same <no-null-cycle-validation> invocation, the same PIVOT-to-exit verdict. The auditor explicitly noted that the verdict had now converged across two consecutive null cycles, confirming the structural posture. The recurrence stabilizes the exit pattern: any subsequent firing should produce the same verdict, and any “new scope” pivot would force scope creep into namespaces this clone is forbidden to touch (sibling Wave 1 staging, Barrier 1 reconciliation, root deliverables, Wave 2 enrichment).

3.4 Boundary discipline

Across all three cycles, the clone refrained from:

- modifying the frozen schema (Barrier 0 invariant);
- writing into sibling clones' staging namespaces;
- performing the Barrier 1 reconciliation work (LQE × PBDB taxon dedup, plant-binomial crosswalk against the M1.1 backbone) — this is reserved for the Barrier 1 coordinator;
- inferring anachronism edges beyond what the cited literature asserts.

4. Discussion

4.1 What is new

The new artifacts contributed by this clone:

1. **A reproducible paleobotany-and-megafauna staging layer** for PhytoGraph Track 2, structured for clean merge into the shared substrate at Barrier 1.
2. **A Janzen-Martin canon of 31 literature-cited anachronism edges** explicitly reserved as a held-out validation set — a precondition for an honest Track 2 evaluation.
3. **A verbatim-confidence convention** that preserves source-stated uncertainty (PBDB intervals, LQE flags, Faurby & Svenning `range_type_code`) rather than collapsing it to a single internal scale. This is necessary because Track 2's central question depends on the difference between reconstructed Late Pleistocene range and anthropogenically-shrunk modern range, and that distinction lives entirely in source-provided fields.

4.2 What is integrated

PBDB, LQE, Faurby & Svenning, and IUCN are all pre-existing datasets. The staging tables integrate them under one schema; they do not produce new biological claims.

4.3 What is deferred to Barrier 1

- LQE × PBDB taxon-name deduplication.
- Plant-binomial stub crosswalk against the M1.1 taxonomy backbone (clone 1's product).
- Live re-ingest of IUCN polygons (network was unavailable; canonical keys preserved).
- Any modification to the shared substrate.

4.4 What is deferred to Wave 2 / Track 2 enrichment (M2.T2)

- Inferred `anachronism_candidate_edges` (this clone staged only literature-asserted edges; M2.T2 is where the inference instrument runs).
- Held-out validation against the 31-edge canon.

4.5 The null-cycle posture

Two consecutive null cycles, each producing the same PIVOT-to-exit verdict, demonstrate convergence on the framework-prescribed terminal posture. The clone is not idle by accident; it is idle because its scope is exhausted and the directive forbids scope creep. The remaining termination mechanism is the harness low-output detector. This is a documented, expected, framework-conformant state.

5. Open Questions and Handoff Notes

For the Barrier 1 coordinator:

- The clone's staging tables are dedup-ready against canonical keys (taxon-ID + edge-type + supporting-source-set per the directive's deduplication discipline). Verify before merging into the shared substrate.
- Re-run the IUCN polygon download under the preserved query keys when live network is available.
- Cross-reference LQE and PBDB extinct-fauna nodes; expect overlap on canonical Pleistocene megafauna.

For the M2.T2 (Ghost-Partner Ranker) clone:

- The 31 canon edges in `anachronism_canon/anachronism_candidate_edges.jsonl` are held-out. Declare no-leakage upfront. The model must not see these labels during training; its job is to rediscover them.

For the campaign auditor:

- The cycle-1 manager intervention's `_manager/ledger-integrity action_required`, legacy plant-taxonomy ledger errors, and root-file `org_check` warnings on cycle-1 deliverables are carry-over items not owned by this clone. They are documented here to prevent re-attribution.

Appendix: Implementation Details

A.1 File inventory (workspace-relative)

substrate/staging/paleobotany_sources/	
INGEST_AUDIT.md	(required artifact, 321 lines)
coverage_row.json	
_lib/	shared schema helpers
tests/	10 schema-conformance tests, all passing
QUARANTINE/	empty (0 non-conformant records)
pbdb/	
build_pbdb.py	
extinct_fauna.jsonl	
paleo_context.jsonl	
raw/	
lqe/	
build_lqe.py	
extinct_fauna.jsonl	
paleo_context.jsonl	
source_citations.jsonl	
faurby_svenning/	
build_faurby_svenning.py	
taxon_nodes.jsonl	
region_nodes.jsonl	
distribution_edges.jsonl	
ranges/	
iucn/	scaffold (live polygon download deferred)
anachronism_canon/	
build_anachronism_canon.py	
anachronism_candidate_edges.jsonl	31 literature-cited edges, 0 inferred

taxon_stubs.jsonl
fruit_type_stubs.jsonl
SOURCE_PAIRS.md

A.2 Test results

10/10 schema-conformance tests passing as of cycle 1. State unchanged through cycles 2 and 3 (zero mutations).

A.3 Cycle session references

Cycle	Role	Session ID
1	researcher	09c22174-7c20-4dc7-80bf-b6f6e6ef233e
1	worker	f4a5f3f0-092a-4b7e-9e67-92f5350ab9ca
1	auditor	821fb9c2-2c54-412b-ad56-01a752ab8fda
2	researcher	1044311e-61cb-4602-85e7-bdf832c53982
2	worker	ecda6cac-44b0-4cfc-88d1-8da4bf8ddf7b
2	auditor	a2400a23-fdab-4b44-8436-3ec5dce402e0
3	researcher	4b268fd6-1b80-466e-88aa-e27eff9d6e84
3	worker	ff414638-f1cf-4e16-a9a8-4cb65b6917a8
3	auditor	52cd95b0-bdd6-41df-b7ec-48e63100da1e

A.4 Cross-reference map

- Required output artifact: `substrate/staging/paleobotany_sources/INGEST_AUDIT.md` — present.
- Conductor handoff: `/home/jeremyad55/workspace/.long-exposure/fork-e34b5b2c1c6c/clone-2/merge_report.md` — present, unchanged since cycle 1.
- Held-out validation set for downstream M2.T2 / M3.T2: `anachronism_canon/anachronism_candidate_edges.jsonl` (31 rows).
- Wave 1 M1.4 milestone: VALIDATED at cycle 1; status unchanged through cycles 2 and 3.

A.5 Verdict rationale

M1.4's required deliverables were produced and validated in cycle 1. Cycles 2 and 3 correctly held the boundary against scope creep and converged on the framework-prescribed PIVOT-to-exit posture. The clone's scoped objective is complete. The merge verdict reflects completion of the assigned milestone, not the null-cycle posture of the trailing cycles — those are evidence of disciplined exit, not of incomplete work.

validated [END OUTPUT: report]